

I Leverage

h_{ii}

$H_{n \times n}$

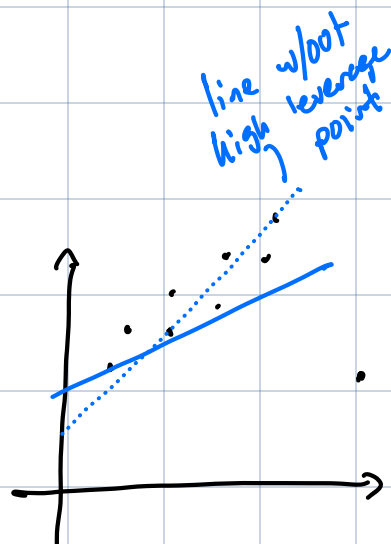
Properties of h_{ii}

$$① \quad 0 \leq h_{ii} \leq 1$$

$$② \quad \sum_i h_{ii} = \text{tr}(H) = p \quad \bar{h} = \frac{p}{n}$$

$$\text{High leverage} > 3 \frac{p}{n}$$

$$p = 10 \quad \frac{30}{100}$$



II Outliers

Externally studentized residual

$$\text{stud}(e_i) = \frac{e_i}{\sqrt{MSE_{(i)}(1-h_{ii})}} \sim t_{(n-1)-p}$$

(Handwritten pink annotations: an arrow points from e_i to $y_i - \hat{y}_i = 10$; another arrow points from $(n-1)-p$ to 2; a third arrow points from $1-h_{ii}$ to leverage)

$$MSE_{(i)} = \frac{\hat{\sigma}_{(i)}^2}{(n-1)-p} = \frac{SSE_{(i)}}{(n-1)-p} = \frac{\sum_j e_j^2}{(n-1)-p}$$

because the model is fit on $n-1$ points

MSE_(i) from model fit on all obs. except i^{th} point.

$$\text{Var}(e) = \sigma^2(I-H)$$

$$\hat{SE}(e) = \sqrt{\hat{\sigma}^2(I-H)}$$

$$\Rightarrow \hat{SE}(e_i) = \sqrt{\hat{\sigma}_{(i)}^2(1-h_{ii})}$$

leave out i^{th}
point
(externally)

$$|\text{stud}(e_i)| > t_{n-1-p, (1-\alpha/2)}^* \Rightarrow i^{\text{th}} \text{ point is an outlier}$$

III Influential Points

High leverage + outliers

Cook's Distance

$$D_i = \frac{(y_i - \hat{y}_i)^2}{p \hat{\sigma}^2} \frac{h_{ii}}{(1-h_{ii})^2}$$

Outlier
part

leverage
part

$$\begin{array}{l} h_{ii} \\ \frac{h_{ii}}{(1-h_{ii})^2} \end{array}$$

$$D_i > \frac{4}{n} \Rightarrow \text{"Influential"}$$

What to do?

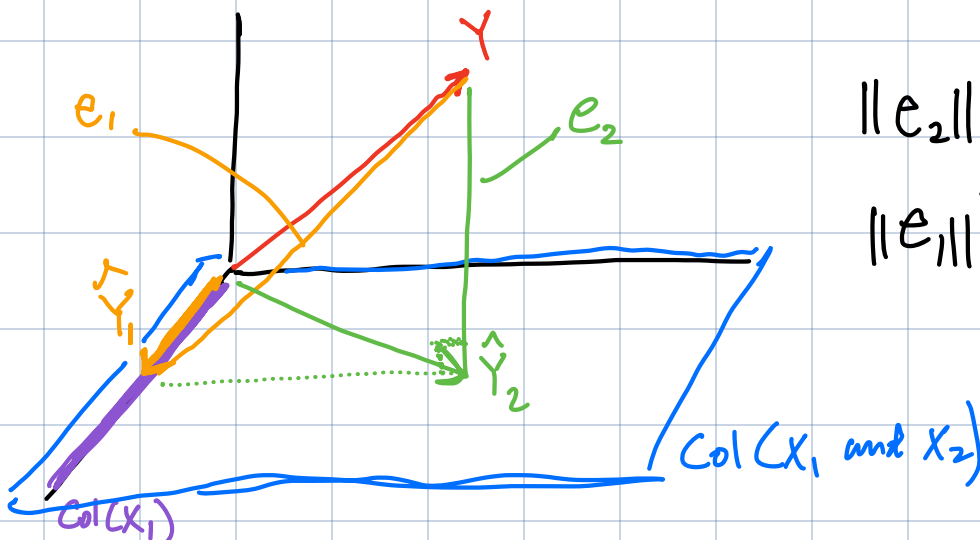
- Do not delete unless it's erroneous.
- If not sure, analyze data twice, with and w/out. Report both sets of results.
- Try different model (new data, different method)

IV Model Selection

1. Underspecified - missing important predictors, biased estimates
2. Extraneous predictors - unbiased estimates, not too bad
3. Overspecified - redundant predictors (multicollinearity)
4. Correctly specified

V Problem w/ R^2

- As you add predictors, $R^2 \uparrow$



$$\|e_2\| \leq \|e_1\|$$

$$\|e_1\|^2 = \|e_2\|^2 + \|\hat{Y}_2 - \hat{Y}_1\|^2$$

We'll show that 19 junk predictors $\Rightarrow R^2 \approx .79$

$$\text{adj } R^2 = 1 - \frac{\text{SSE}/(n-p)}{\text{SST}/(n-1)} \quad \checkmark \text{ this won't happen}$$

$\text{adj } R^2$ still increases for insignificant predictors
($p\text{-val} < .32$). $\text{adj } R^2$ overfits, adds
too many predictors.

VI Mallows' C_p

2 issues with models:

- ① Bias in predictions
- ② Variance in predictions

Expected (standardized) Prediction error

$$\begin{aligned} \Gamma_p &= \frac{1}{\sigma^2} \sum_i E(\hat{y}_i - E(y_i))^2 \\ &= \frac{1}{\sigma^2} \left[\underbrace{\sum_i \text{Var}(\hat{y}_i)}_{\text{Variance (a)}} + \underbrace{\sum_i (E(\hat{y}_i) - E(y_i))^2}_{\text{bias}^2 (b)} \right] \end{aligned}$$

true mean response

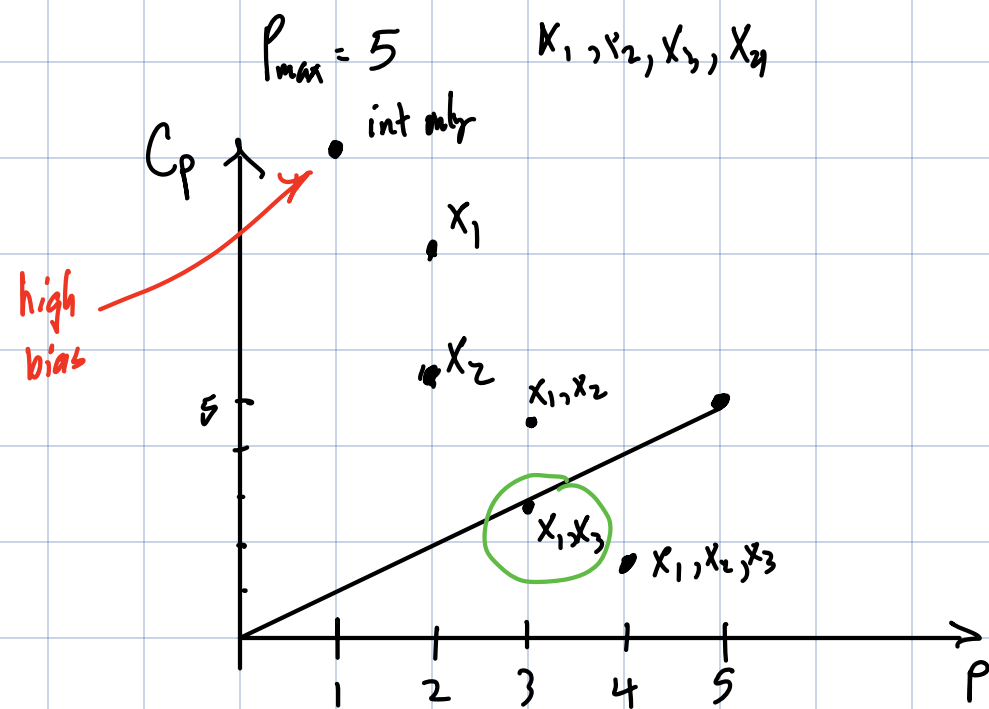
$$(a) \text{Var}(\hat{Y}) = \sigma^2 H \Rightarrow \sum_i \text{Var}(\hat{y}_i) = \sigma^2 \text{tr}(H) = \sigma^2 p$$

$$(b) E(SSE_p) - (n-p)\sigma^2$$

$$\frac{1}{\sigma^2} \left[p\sigma^2 + E(SSE_p) - (n-p)\sigma^2 \right]$$

$$C_p = \frac{SSE_p}{\hat{\sigma}_F^2} - (n-2p)$$

$$\text{Full model} \Rightarrow C_{p_{\max}} = \frac{SSE_F}{SSE_F/(n-p_{\max})} - (n-2p_{\max}) = p_{\max}$$



smallest p s.t. C_p below or on the line $\Rightarrow$ low bias

Parsimonious

VII

AIC

Akaike Information Criterion

$$\varepsilon \sim N(0, \sigma^2 I)$$

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \|Y - X\beta\|_2^2$$

$$-2\ell(\hat{\beta}, \hat{\sigma}^2) = n \ln SSE_p - n \ln n + n \ln 2\pi + n$$

$$AIC = -2\ell(\hat{\beta}, \hat{\sigma}^2) + 2(p+1)$$